Contact Management System

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OUTLINE

- Abstract
- Existing System
- Proposed System
- System Architecture
- Objectives
- Co’s Attainments – FWD
- Co’s Attainments – DSA
- Screen Shots for FWD & DSA
- Conclusion
- Geo-Tagged Photos

Abstract

- The Contact Management System is a web-based application developed to store and manage contact information efficiently. The system allows users to add, search, sort, and display contact details through a simple and user-friendly interface. It is built using HTML, CSS, and JavaScript, which provide the structure, design, and interactivity of the application.
- The system stores contact details such as name and phone number and performs operations like searching and sorting using basic algorithmic techniques. JavaScript is used to handle user input, perform validations, and dynamically update the contact list using DOM manipulation.
- This project demonstrates important concepts from the Forward Web Development (FWD) course, including HTML forms, CSS styling and layout, JavaScript fundamentals, event handling, and dynamic web page updates. It shows how web technologies can be used to develop interactive applications for managing and organizing data efficiently.

Existing System

- In the existing system, contact information is commonly stored manually in notebooks, diaries, or basic phone contact lists. These traditional methods are simple but often inefficient for managing a large number of contacts. When contacts are stored manually, it becomes difficult to maintain and organize the information properly.
- Searching for a specific contact requires going through the records one by one, which takes time and may lead to mistakes. Updating or modifying contact details can also be inconvenient, as it may require rewriting the information. In addition, arranging contacts in alphabetical order or managing them systematically is not easily possible using manual methods.
- Because of these limitations, the existing system lacks efficient searching, sorting, and data management capabilities. This creates the need for a digital system that can store and organize contact information in a faster and more efficient way.

Proposed System

- The proposed system is a Contact Management System designed to store and manage contact information in a more efficient and organized way. The system allows users to add, search, sort, and display contact details through a simple and user-friendly interface. By using a digital platform, contact information can be managed quickly and accurately.
- In this system, contact details such as name and phone number are stored and managed using programming logic and basic data structure concepts. The system provides features like searching for a contact easily, sorting contacts alphabetically, and displaying the stored contacts in an organized format. Input validation is also implemented to ensure correct data entry.
- The proposed system reduces the difficulties of manual contact management and improves efficiency, accuracy, and speed. By using web technologies such as HTML, CSS, and JavaScript, the system provides an interactive and easy-to-use solution for managing contact information effectively.

System Architecture

- The Contact Management System follows a simple client-side architecture where the user interacts with the application through a web interface. The system is built using HTML for the structure, CSS for styling, and JavaScript for functionality and interactivity.
- In this architecture, the user enters contact details such as name and phone number through input fields in the interface. JavaScript processes the user input, performs validation, and executes operations like adding, searching, sorting, and displaying contacts. The contact data is temporarily stored in an array structure within the application.
- The processed data is then dynamically displayed on the web page using DOM manipulation. This architecture allows the system to handle user interactions efficiently and update the contact list instantly without reloading the page. The overall design ensures a simple, interactive, and efficient way to manage contact information.

Objectives of the Project

- To develop a simple and efficient **Contact Management System** for storing and managing contact information.
- To allow users to **add, search, sort, and display contacts** easily.
- To implement **basic data management operations** using programming logic and algorithms.
- To apply **web development technologies such as HTML, CSS, and JavaScript** in building an interactive application.
- To demonstrate concepts like **input validation, DOM manipulation, and dynamic data handling**.
- To provide a **user-friendly interface** for organizing and accessing contact details efficiently.

CO Attainment of FWD:

- CO1:
- Applied the fundamentals of HTML and CSS to create the structure and styling of the Contact Management System interface.
- CO2:
- Implemented form elements and input fields to collect user data and applied CSS layout techniques to design a clean and organized user interface.
- CO3:
- Used JavaScript programming concepts such as variables, functions, conditions, arrays, and objects to manage contact data.
- CO4:
- Implemented DOM manipulation and event handling to perform operations like adding, searching, sorting, and displaying contacts dynamically.
- CO5:
- Applied JavaScript-based validation and dynamic user input handling to ensure correct data entry and efficient contact management.

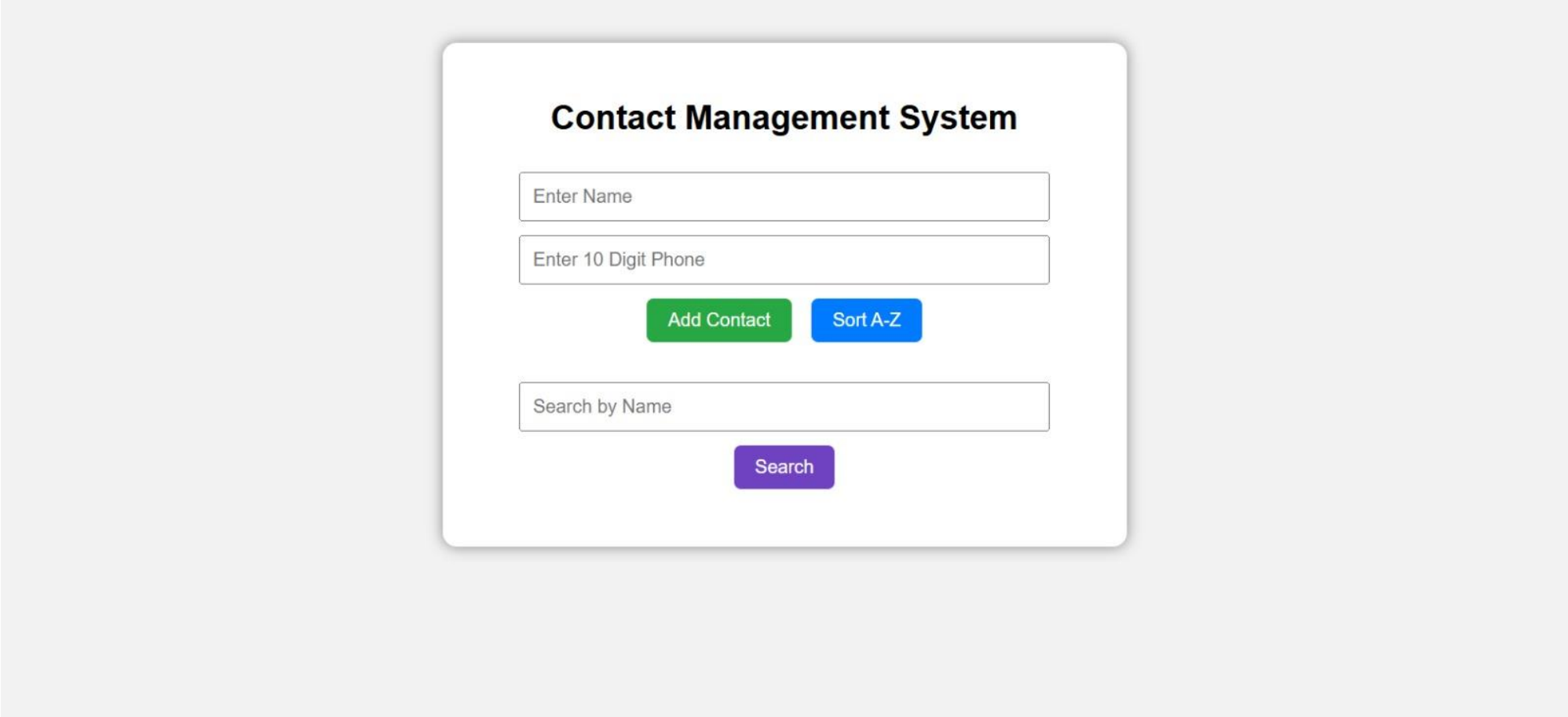
CO Attainment of DSA

- CO1:
- Implemented searching and sorting techniques such as linear search and alphabetical sorting to manage and organize contact data efficiently.
- CO2:
- Used the Array Abstract Data Type (ADT) to store contact details and performed operations such as insertion, searching, and traversal.
- CO3:
- Applied the concept of sequential data processing using array-based storage to manage contact information.
- CO4:
- Utilized collection-like data structures to store and retrieve contact information efficiently through lookup operations.
- CO5:
- Developed a practical application using linear data structures to store, manage, and retrieve contact details.
- CO6:
- Designed and implemented a working application that demonstrates the use of data structures and algorithms to solve a real-world problem.

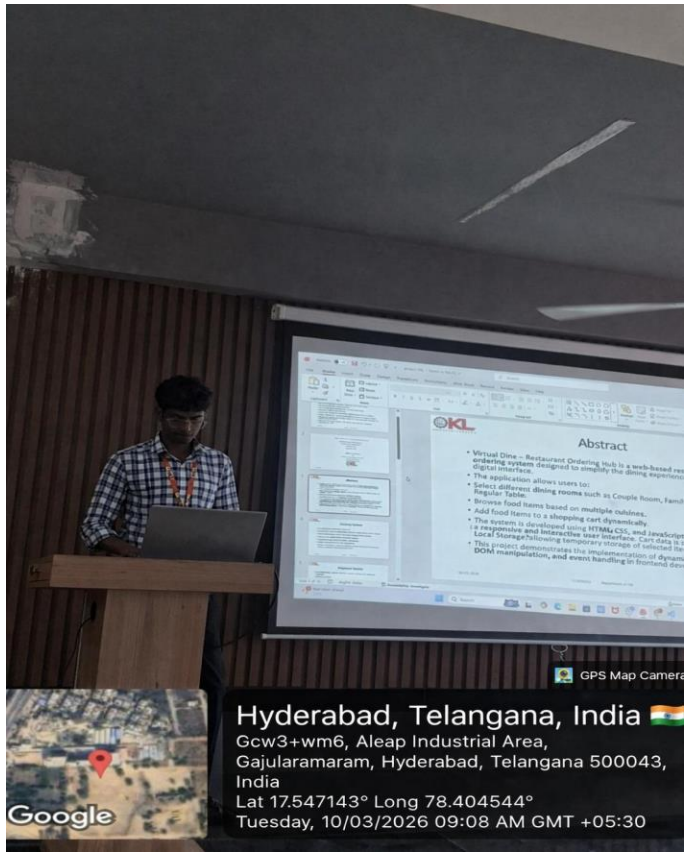
Conclusion

- The Contact Management System was successfully developed to efficiently store and manage contact information. The system allows users to add, search, sort, and display contact details through a simple and interactive interface. By using web technologies such as HTML, CSS, and JavaScript, the application provides a user-friendly way to organize and access contact data.
- This project also demonstrates the practical application of concepts from both Forward Web Development (FWD) and Data Structures and Algorithms (DSA), such as form handling, DOM manipulation, arrays, searching, and sorting techniques. Overall, the system improves the efficiency of managing contacts and shows how web technologies and algorithms can be combined to develop useful real-world applications.

Screenshot


Contact Management System
Enter Name
Enter 10 Digit Phone
Add Contact
Sort A-Z
Search by Name
Search

Geo-Tagged Photos



Thank you